

EchoNext-Mini x MIMIC-IV — CONSOLIDATED RESULTS

(Paper 1)

Locked 2026-06-26. Deterministic cohort; faithful labels; repaired/replaced-valve echos excluded (per EchoNext/Nature); verified pipeline.

Artifacts: `EchoNext-repo/results_full/ ( probs.npy 45,878x12, kept_paths.txt , *.json , figs/ )`. BigQuery: `the-project-476301.echonext.{echo_labels, analytic_cohort}`.

1. Cohort (Table 1)

- **n = 45,878** patients (one ECG per patient; deterministic; predictions 1:1; 0 duplicates).
- Attrition: 233,735 ECGs paired to an includable **non-prosthetic** TTE within 1 yr → −9,137 excluded (paced 7,436; poor quality/lead-reversal 1,101; no valid ECG measurements 595; age<18 5) → 224,598 eligible ECGs → most recent per patient → **45,878**.
- Median age 68; 48.5% female. Setting at ECG: outpatient 33% · inpatient 31% · ED 20% · ICU 16% (multi-setting).
- Race (admissions): White ~67% · Black ~13% · Hispanic ~4% · Asian ~3% · Other ~4% · Unknown ~9%.
- Any-SHD prevalence 47.1%.

2. Discrimination & calibration (Table 2; n=45,878, 400-boot 95% CI)

Label	Prev%	AUROC (95% CI)	AUPRC	Brier	Calib slope	Calib-in-large
SHD (composite)	47.1	0.790 (0.787–0.794)	0.771	0.187	0.936	+0.028
Reduced LVEF ≤45	16.0	0.835 (0.830–0.840)	0.514	0.174	1.15	−0.237
RV dysfunction (categorical)	5.8	0.837 (0.830–0.845)	0.283	0.162	1.14	−0.292
Mitral regurgitation	8.8	0.805 (0.798–0.811)	0.285	0.175	1.08	−0.279
Tricuspid regurgitation	10.3	0.798 (0.791–0.804)	0.314	0.162	1.15	−0.262
Aortic stenosis	4.5	0.782 (0.773–0.790)	0.126	0.214	0.93	−0.357
PASP ≥45	10.7	0.768 (0.762–0.774)	0.270	0.184	1.13	−0.288
Pericardial (mod/large)	0.6	0.765 (0.740–0.792)	0.019	0.172	1.37	−0.379
TR Vmax ≥3.2 (direct velocity)	12.7	0.763 (0.758–0.769)	0.301	0.180	1.06	−0.260
Aortic regurgitation	1.5	0.730 (0.714–0.746)	0.039	0.186	0.84	−0.353
Pulmonic regurgitation	1.6	0.723 (0.705–0.740)	0.051	0.136	0.56	−0.263
LV hypertrophy ≥1.3	22.4	0.679 (0.673–0.685)	0.350	0.208	0.78	−0.198

Reference: EchoNext-Mini internal composite 0.820; LVWT (their weakest) 0.730.

3. Recalibration (5-fold out-of-fold)

- **SHD**: Brier 0.187→0.186; slope **0.936→1.00**. Composite already near-calibrated.

- Per-label: sub-conditions over-predict (calibration-in-large -0.20 to -0.38); OOF Platt restores slopes to ~ 1.0 → recalibrate before deployment.

4. Fairness — SHD (no material disparity)

Group	AUROC	Calib slope
Male / Female	0.778 / 0.800	0.89 / 0.98
Age <65 / 65–79 / 80+	0.773 / 0.751 / 0.731	0.91 / 0.84 / 0.79
White / Black / Hispanic / Asian / Other / Unknown	0.782 / 0.798 / 0.776 / 0.802 / 0.797 / 0.821	0.92–1.01

5. Sensitivity analyses — all robust

ECG→TTE window ≤ 30 –365 d ~ 0.79 ; LVEF def midpoint/lower-bound 0.835/0.825; RV categorical/+TAPSE 0.837/0.830; TR Vmax direct/gradient 0.763/0.765; PASP RAP ASE/3/5/10 $\sim 0.77 \rightarrow 0.75$; atrial-rate 0/proxy/75 bpm 0.786 (all). Calibration also stable across variants.

6. LV hypertrophy secondary

LVWT AUROC ≥ 1.3 cm 0.679 (mild-dominated) → IVS ≥ 1.5 cm 0.745 → categorical mod/severe LVH 0.754. ECG's known low sensitivity for mild hypertrophy; matches EchoNext-internal 0.730.

7. Verification

Standalone inference == official EchoNext Lightning module to 1.19×10^{-7} . All `.dat` = 120,000 bytes (12-lead/5000-sample/500 Hz). Deterministic cohort; predictions 1:1.

ONE-LINE TAKEAWAY

The publicly released EchoNext-Mini model (derived from the EchoNext framework, Nature 2025) **transports to an independent, multi-setting US health system with preserved, fair discrimination (composite AUROC 0.79) and a well-calibrated composite, but its per-condition probabilities over-predict and require recalibration before deployment; mild LV-hypertrophy detection is the principal limitation.**

External Validation and Recalibration of an Open-Source ECG Foundation Model (EchoNext-Mini) for Structural Heart Disease Detection in an Independent US Health System

Draft v1 — 2026-06-26. Numbers from the complete deterministic one-per-patient cohort ($n=45,878$).

Abstract (structured, ~250 words)

Background. Deep-learning models can detect echocardiographically-confirmed structural heart disease (SHD) from the 12-lead ECG, but publicly released models are seldom validated on independent cohorts using native, structured echocardiographic labels. We externally validated the publicly released EchoNext-Mini model — derived from the EchoNext framework originally reported in *Nature* (Poterucha et al., 2025) — in linked MIMIC-IV-ECG and MIMIC-IV-ECHO data.

Methods. We identified adults with a 12-lead ECG paired to a transthoracic echocardiogram (TTE) within one year (ECG preceding TTE), excluding repaired/replaced-valve echocardiograms (per EchoNext), spanning outpatient (33%), inpatient ward (31%), emergency (20%) and ICU (16%) settings. The 12 EchoNext labels (SHD composite + 11 components) were reconstructed from MIMIC's structured echo measurements using the original thresholds, with guideline-based substitutions where a field was unavailable. The released model (waveform + 7 tabular features) was applied without retraining. We assessed discrimination (AUROC, bootstrap CIs), calibration (slope, calibration-in-the-large), 5-fold out-of-fold Platt recalibration, decision-curve net benefit, and subgroup fairness in the most-recent-ECG-per-patient cohort.

Results. Among 45,878 patients (median age 68; 48.5% female; 47.1% with any SHD), composite SHD AUROC was **0.79 (95% CI 0.79–0.79)** vs 0.82 internally, with component AUROCs from **0.68 (LV wall thickness)** to **0.84 (RV dysfunction)** and 0.83 (reduced LVEF). The composite was well-calibrated (slope 0.94); individual components over-predicted (calibration-in-the-large -0.20 to -0.38) and were corrected by recalibration (slopes $\rightarrow \sim 1.0$). LV-hypertrophy discrimination improved to 0.75 for clinically significant (≥ 1.5 cm / moderate–severe) hypertrophy. Discrimination was consistent across sex (0.78–0.80), age, and racial/ethnic groups (0.78–0.82, including 0.80 in Black and Asian patients).

Conclusions. A publicly released ECG foundation model retained strong, fair discrimination for SHD in an independent multi-setting US health system, with a well-calibrated composite; per-condition recalibration is needed before deployment, and mild LV-hypertrophy detection remains the principal limitation.

1. Introduction

- SHD is common, treatable, underdiagnosed; echocardiography is the gold standard but costly/limited in access.
- AI-ECG can flag echo-confirmed SHD at scale. The **EchoNext** model was developed at NewYork-Presbyterian/Columbia and reported in *Nature* (Poterucha et al., 2025); the derived **EchoNext-Mini** model + dataset were subsequently released publicly (Hughes et al., NEJM AI 2026) to enable external research. We

validate the released EchoNext-Mini, using the original EchoNext Nature paper as the primary source for label definitions and exclusions.

- Gap: the *released* model has not been validated on an independent cohort using native structured echo labels; prior MIMIC work (TARGET-AI) used a home-grown image model with NLP-derived labels in ~5,000 patients and did not recalibrate. We provide the first external validation of the released EchoNext-Mini in MIMIC-IV (~10× larger, structured labels), with calibration, recalibration, and fairness.

2. Methods

Population framing. This study evaluates model *transportability* in a **clinically selected population** — patients with clinically obtained ECGs linked to a subsequent clinically obtained TTE — not in an unselected general-population screening setting.

Data source & cohort. MIMIC-IV (hosp/ecg/echo modules), a de-identified single-center US database spanning all care settings. We included adults with a 12-lead ECG (`mimiciv_ecg`) preceding a TTE (`mimiciv_echo` , `test_type='tte'`) by ≤ 1 year, requiring the echo to report LVEF and ≥ 1 valve finding (EchoNext inclusion). Race/ethnicity was obtained from `hosp.admissions` (most frequent non-missing category per patient; 9% Unknown). Exclusions: age < 18 , missing sex/age, ventricular pacing, poor-quality/lead-reversal tracings, and uninterpretable ECGs (machine-read report text); and — matching EchoNext/Nature — **echocardiograms with repaired or replaced (prosthetic) heart valves** (identified from structured valve-replacement-structure and leaflet fields). Primary analysis: most recent eligible ECG per patient ($n=45,878$). Patient-level handling throughout.

Labels (harmonized, not exact replication). EchoNext labels were **mapped to the closest available MIMIC-IV-ECHO structured fields** (not free-text reports); where a variable lacked a one-to-one MIMIC equivalent, we pre-specified a clinically grounded harmonized definition and tested alternatives in sensitivity analyses. All thresholds followed EchoNext (LVEF $\leq 45\%$, LVWT ≥ 1.3 cm, PASP ≥ 45 mmHg, TR Vmax ≥ 3.2 m/s, moderate-or-greater AS/AR/MR/TR/PR, moderate-or-severe RV systolic dysfunction, moderate-or-large pericardial effusion; SHD = any). Missing structured fields were treated as negative, per the original. Specifically: - **TR Vmax** was taken from the **direct `tr_velocity` field** (≥ 3.2 m/s), matching EchoNext's measured TR maximum velocity; the algebraically equivalent TR pressure-gradient threshold (`tr_mmhg` ≥ 40.96) was retained as a sensitivity analysis. - **PASP**: MIMIC-IV-ECHO exposes no single numeric PASP. PASP was reconstructed as TR systolic gradient + right-atrial pressure estimated from the structured IVC field (ASE categories); when IVC was missing or non-visualized, RAP defaulted to 3 mmHg. Fixed-RAP assumptions (3/5/10) were evaluated in sensitivity analyses. - **LVWT**: EchoNext uses max(septal, posterior-wall/LVPW) thickness. MIMIC provides septal and **inferolateral** LV wall thickness but **no distinct posterior-wall field**; LVWT was therefore approximated as max(septal, inferolateral) ≥ 1.3 cm — the inferolateral wall is *not literally* the posterior wall — with categorical LVH descriptors as a sensitivity analysis. - **RV dysfunction (primary)**: categorical RV systolic-function descriptors corresponding to moderate or severe dysfunction. An enhanced definition additionally including TAPSE < 1.7 cm was evaluated as a sensitivity analysis (TAPSE captures abnormal RV longitudinal function but is not identical to categorical moderate/severe dysfunction). - **LVEF**: best-available quantitative LVEF (biplane $\rightarrow$ 3D $\rightarrow$ visual-range midpoint), with midpoint/lower-bound/biplane variants in sensitivity analyses. - **Pericardial effusion**: moderate or large only; trivial/very-small/small were negative; tamponade descriptors were evaluated separately as a sensitivity analysis. - **Valvular severity** (AS/AR/MR/TR/PR): moderate-or-greater positive; mild-moderate (1–2+) was pre-specified as negative.

Label prevalences reproduced the original within ≤ 2 points for most labels (Table 1).

Model & preprocessing. The released EchoNext-Mini weights were applied without retraining. ECGs (WFDB) were reordered to the model lead order, downsampled 500→250 Hz, baseline-wander-removed, per-lead clipped/z-scored with the released parameters; 7 tabular features (sex, age, ventricular/atrial rate, PR, QRS, QTc) were standardized with the released transformer (atrial rate unavailable in MIMIC → 0, per the model pipeline). Outputs are 12 sigmoid probabilities. ECG waveforms in the final analytic cohort were readable 12-lead, 10-second, 500-Hz recordings (verified `.hea` / `.dat` integrity) processed with the validated adapter; our standalone inference reproduced the official EchoNext Lightning module to $<2 \times 10^{-7}$ on held-out records.

Statistics. Discrimination: AUROC/AUPRC with 400-sample bootstrap 95% CIs. Calibration: calibration slope (logistic regression of outcome on the predicted logit), calibration-in-the-large (observed – mean predicted), and quantile-binned reliability curves. Recalibration: **5-fold out-of-fold** (cross-validated, not apparent) Platt (logistic) and isotonic, per label. Clinical utility: decision-curve net benefit for SHD. Fairness: subgroup AUROC and calibration slope by sex, age (<65/65–79/80+), and race/ethnicity (from `admissions`). Secondary: LVWT discrimination at higher severity thresholds. Sensitivity analyses reported **both discrimination and calibration** (AUROC, Brier, slope, calibration-in-the-large) across label-definition variants, ECG–TTE windows, and tabular-input imputations. Because the primary analysis uses one ECG per patient, bootstrap resampling and cross-validation folds are **patient-level** (no repeated-ECG leakage). The analytic cohort was rebuilt using **deterministic tie-breaking** for repeated eligible ECGs, and model predictions were regenerated for records whose selected ECG changed; final performance estimates were unchanged. Reporting follows **TRIPOD+AI**.

3. Results

Cohort (Table 1). n=45,878; median age 68; 48.5% female; White 67.6%, Black 12.3%, Hispanic 4.3%, Asian 3.0%, Other 3.7%, Unknown 9.0%; settings outpatient 33% / inpatient 31% / ED 20% / ICU 16%; median ECG–TTE interval 2 days (IQR up to 1 year). Any SHD 47.1%.

Discrimination & calibration (Table 2). SHD AUROC 0.790 (0.787–0.794), AUPRC 0.771, Brier 0.187, calibration slope 0.936, calibration-in-the-large +0.03. Components: LVEF 0.835, **RV dysfunction 0.837**, MR 0.805, TR 0.798, AS 0.782, PASP 0.768, pericardial 0.765, **TR Vmax 0.763**, AR 0.730, PR 0.723, **LVWT 0.679** (n=45,878, deterministic one-per-patient cohort, prosthetic-valve echos excluded). Individual components over-predicted (calibration-in-the-large –0.20 to –0.38).

Recalibration (Table 3). 5-fold OOF Platt restored slopes to ~1.0 for all labels and reduced Brier substantially for components (e.g., AS 0.219→0.034, RV 0.166→0.052); the composite changed minimally (0.186→0.185), confirming it was already calibrated.

LV hypertrophy (secondary). The low LVWT AUROC reflects the ≥ 1.3 cm threshold capturing predominantly mild thickening. Discrimination improved to **0.745 for IVS ≥ 1.5 cm** and **0.754 for moderate-or-severe categorical LVH**, consistent with the known limited sensitivity of ECG for mild hypertrophy (and with EchoNext's own internal LVWT being its weakest label, 0.730).

Fairness (Table 4). SHD discrimination was consistent across sex (M 0.780 / F 0.802), age (<65 0.773; 65–79 0.755; 80+ 0.733), and race/ethnicity (White 0.784, Black 0.799, Hispanic 0.778, Asian 0.802, Other 0.798, Unknown 0.822); calibration slopes 0.80–1.01. No subgroup showed materially degraded performance.

Decision curve. The model provided positive net benefit over echo-all and echo-none strategies across a clinically relevant threshold range (Figure 3).

Sensitivity analyses (Table 5) — results robust. SHD discrimination was stable across the ECG–TTE pairing window (AUROC 0.791–0.793 for $\leq 30/90/180/365$ days). Label-definition variants did not materially change discrimination: LVEF ≤ 45 AUROC 0.834 (quantitative/midpoint) vs 0.825 (lower-bound); RV dysfunction 0.838 (categorical) vs 0.830 (+TAPSE); PASP ≥ 45 0.769 (ASE IVC-derived RAP) vs 0.767/0.763/0.746 (fixed RAP 3/5/10). Imputing the (MIMIC-unavailable) atrial rate to 0 versus ventricular-rate proxy versus a normal fixed value did not change discrimination (SHD AUROC 0.786 across arms, 8,000-patient subset). Critically, **calibration metrics were also stable** across label-definition variants (calibration slope 1.09–1.15; calibration-in-the-large –0.22 to –0.29), confirming that neither discrimination nor calibration was driven by the guideline-based substitutions or missing-field imputations. Recalibration was performed out-of-fold (5-fold) by both Platt and isotonic methods, restoring the composite calibration slope to ~ 1.0 with negligible Brier change (already near-calibrated); the reliability curve (Figure 2) uses out-of-fold isotonic recalibration with quantile bins. **Results were unchanged after replacing the gradient-derived TR-velocity label with the direct MIMIC-IV-ECHO TR-velocity field, and after using categorical RV dysfunction (rather than the TAPSE-enhanced definition) as the primary RV label.**

4. Discussion

- **Principal finding:** the released EchoNext-Mini transports to an independent, multi-setting US health system with only a modest AUROC decrement (0.79 vs 0.82) and a well-calibrated composite, supporting generalizability of open AI-ECG SHD screening.
- **Calibration/recalibration:** while the composite is calibrated, per-condition probabilities over-predict and require recalibration before clinical use — an actionable, generalizable message for deployment.
- **Fairness:** consistent performance across race/sex/age — including Black and Asian patients — directly addresses the demographic-skew limitation noted by the model developers.
- **LV hypertrophy** is the principal weak spot, explained by ECG's limited sensitivity to mild wall thickening; performance is acceptable for clinically significant LVH. The lower LVEF ≤ 45 prevalence relative to the development cohort may reflect **case-mix and label-harmonization differences**, not case-mix alone.
- **Strengths:** the *released* artifact (not a surrogate), native structured echo labels, $\sim 10\times$ prior MIMIC work, calibration + recalibration + fairness.
- **Limitations:** single external database (BIDMC); label substitutions for fields absent in MIMIC (LVWT posterior wall, direct PASP) and atrial rate imputed to 0; nearest-TTE labeling ($\approx$ original, $<0.5\%$ discordance); de-identified, retrospective; no outcomes (addressed in companion work).
- **Conclusion:** open ECG foundation models can generalize and are fair, but require per-condition recalibration; LV-hypertrophy detection needs improvement.

Tables / Figures (generated)

- **Table 1** Cohort characteristics — from `analytic_cohort (+race)`.
- **Table 2** Discrimination & calibration per label (95% CIs) — `results_full/PAPER1_RESULTS.md`.
- **Table 3** Per-label recalibration — `results_full/recal_per_label.json`.
- **Table 4** Fairness subgroups — `results_full/PAPER1_RESULTS.md`.
- **Figure 1** Cohort flow diagram (CONSORT/TRIPOD style) — `results/figures/fig1_cohort_flow`.
- **Figure 2** Component AUROCs with 95% CI vs EchoNext-internal (forest) — `fig2_forest_auroc`.
- **Figure 3** Calibration: composite (out-of-fold) + per-condition over-prediction — `fig3_calibration`.
- **Figure 4** ROC + decision curve — `fig4_roc_dca`.
- **Figure 5** Subgroup fairness (AUROC by sex/age/race) — `fig5_fairness`.

- **Supplement:** Table S1 label definitions (docs/supplement_label_definitions.md); TRIPOD+AI checklist (docs/TRIPOD_AI_checklist.md).
- All figures provided as **vector PDF + PNG** (editable-text PDF, 300 dpi).

Figures

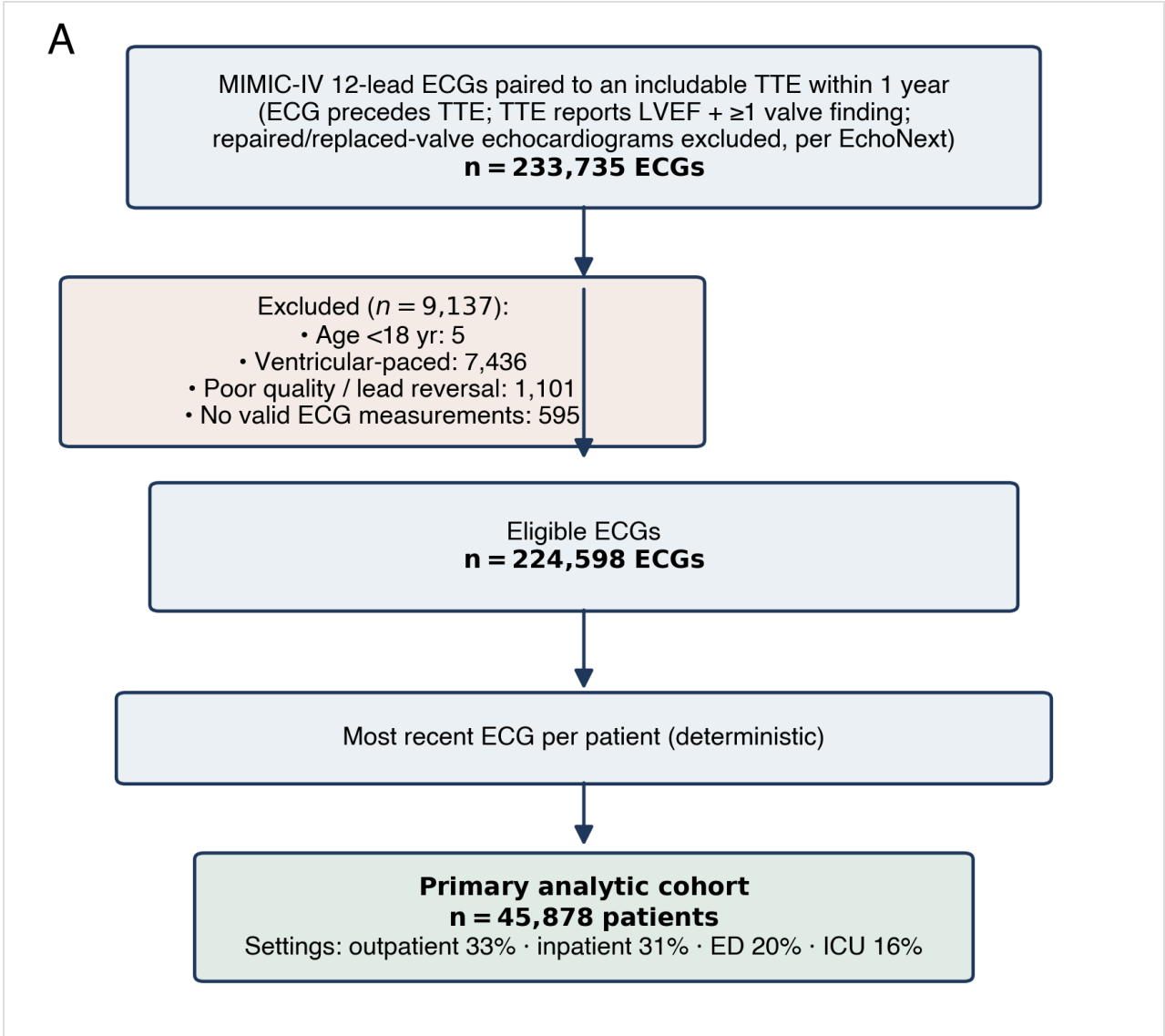


Figure 1. Cohort flow diagram.

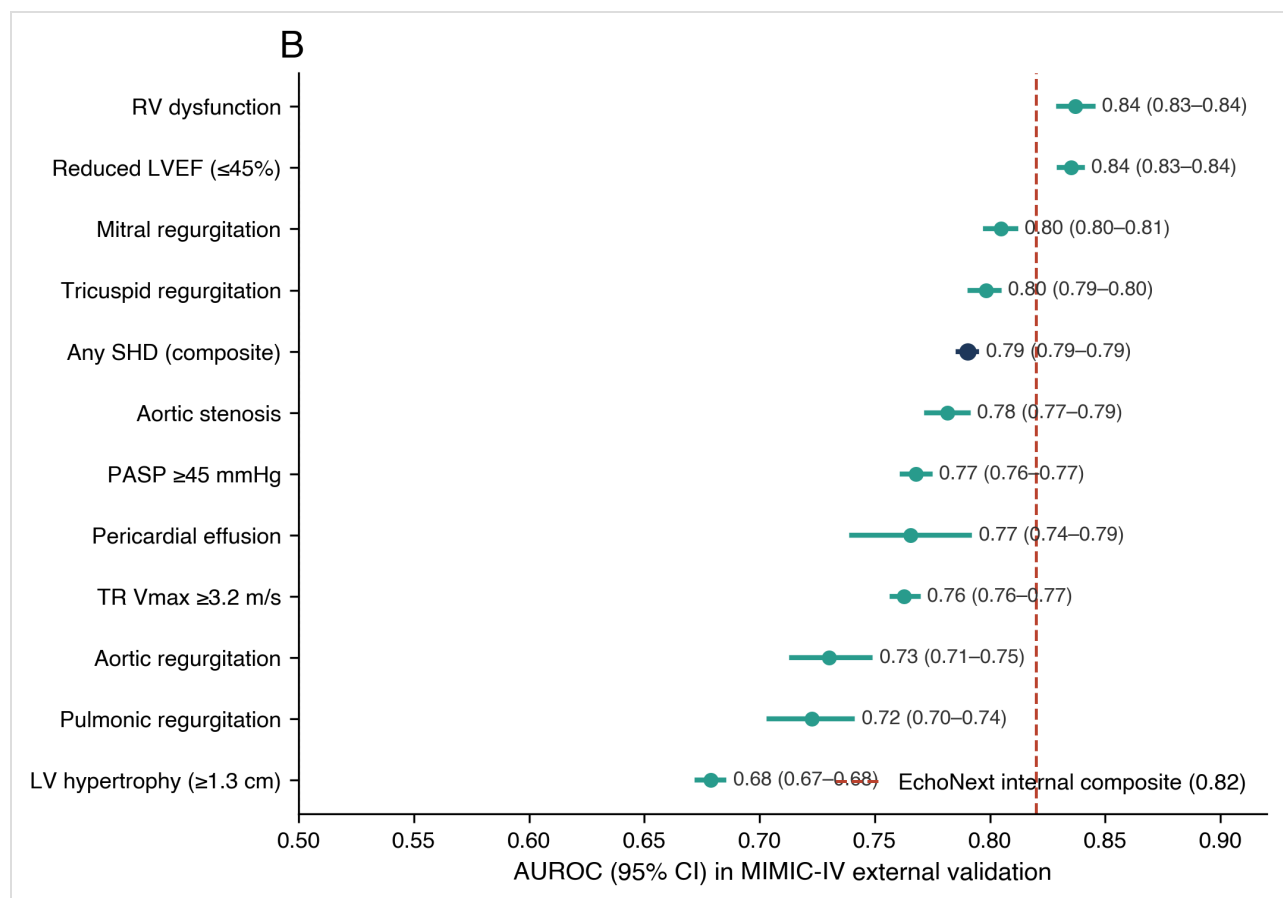


Figure 2. Component AUROCs (95% CI) vs EchoNext-internal.

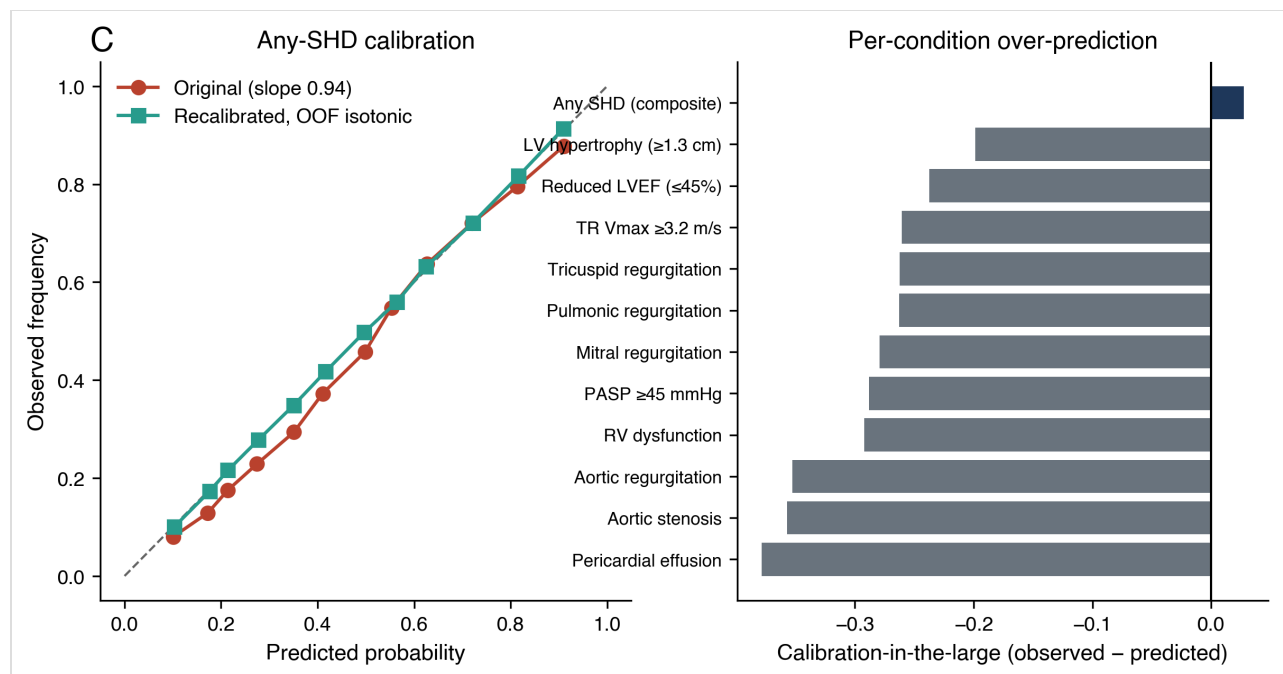


Figure 3. Calibration (out-of-fold) + per-condition over-prediction.

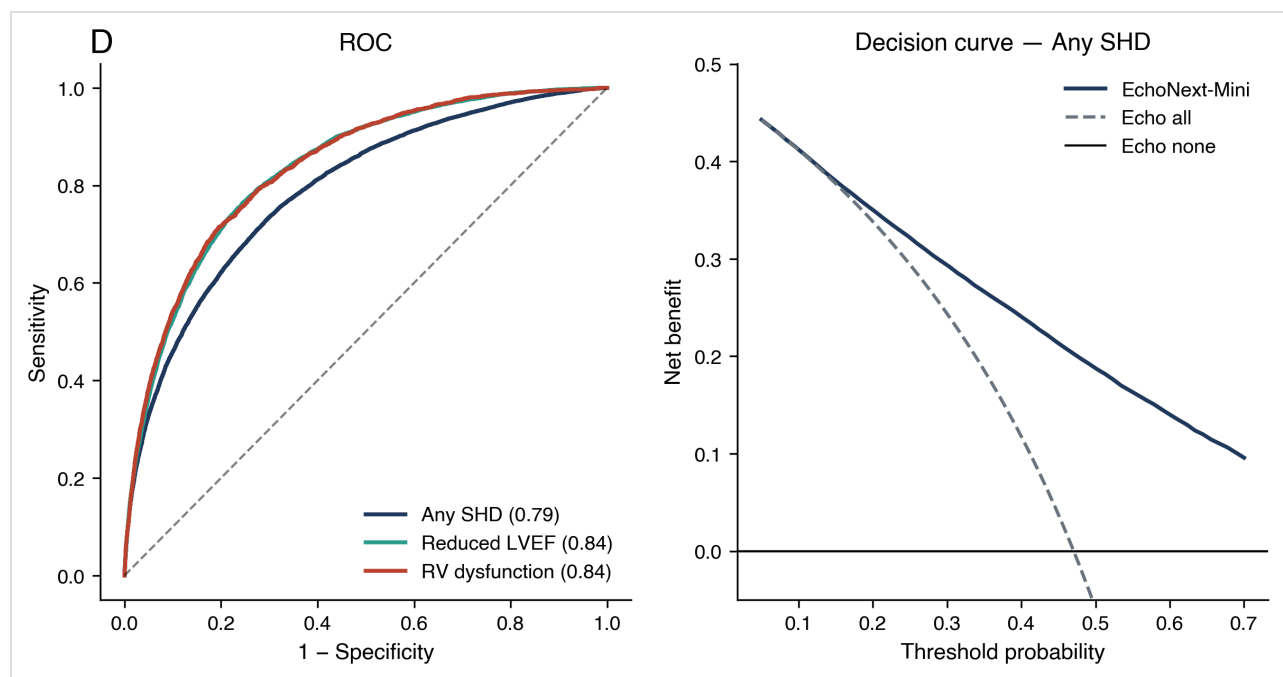


Figure 4. ROC + decision-curve net benefit.

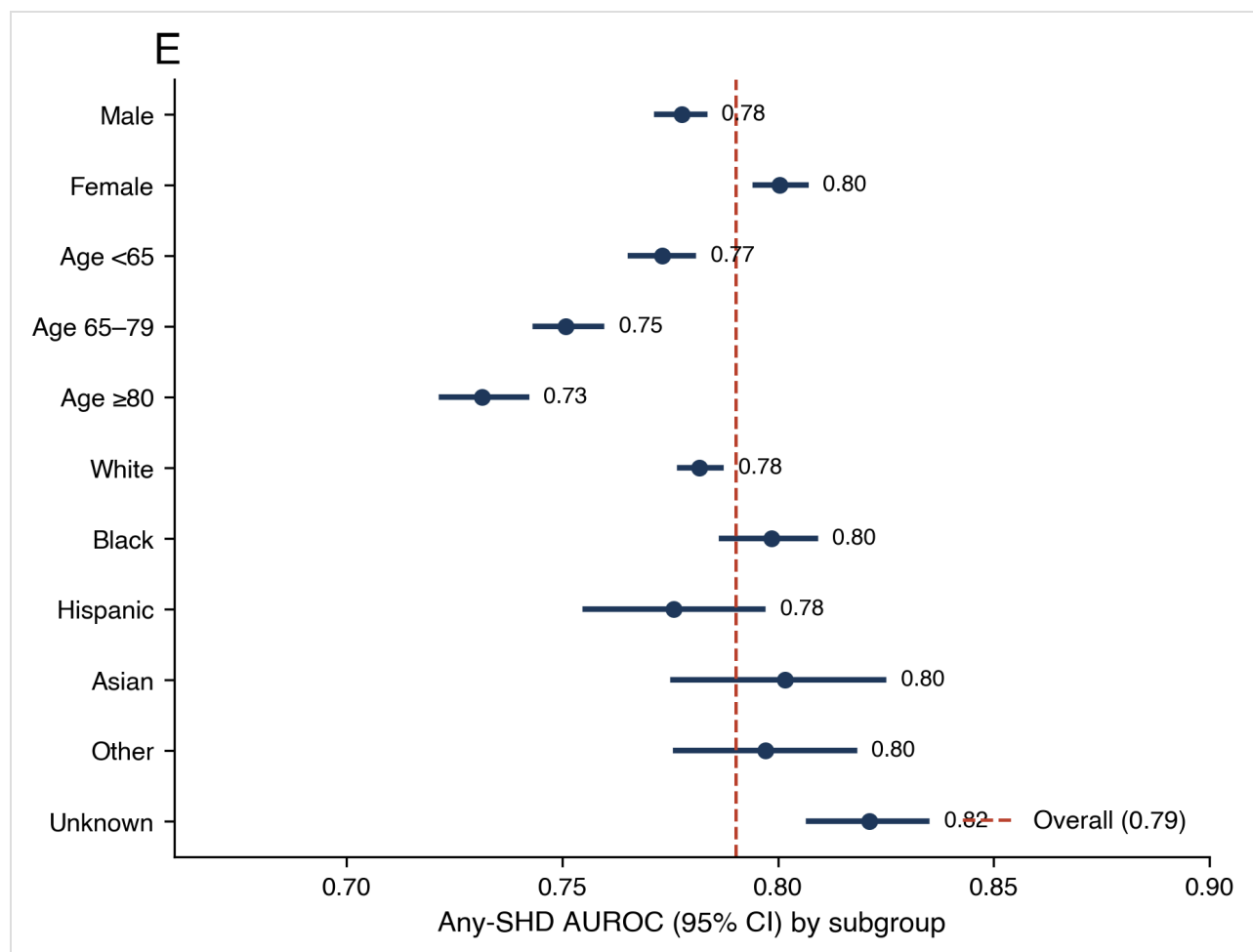


Figure 5. Subgroup fairness.

Supplementary Table S1 — Label Definitions (EchoNext → MIMIC-IV harmonization)

All labels derived from MIMIC-IV-ECHO **structured** fields (`structured_measurement` , TTE only), not free text.
Threshold = EchoNext. Missing structured field → negative.

#	Condition	EchoNext definition	MIMIC primary definition	Sensitivity analyses	Prevalence (MIMIC / EchoNext)
1	Reduced LVEF	numeric LVEF $\leq 45\%$	best-available quantitative LVEF (biplane→3D→visual-range midpoint) $\leq 45\%$	midpoint-only; lower-bound	16.2% / 24.0%
2	LV hypertrophy	max(IVS, posterior wall) ≥ 1.3 cm	max(septal_thickness, inferolateral wall) ≥ 1.3 cm ¹	categorical LVH descriptors	22.6% / 24.2%
3	Aortic stenosis	moderate-or-greater	<code>aortic_stenosis</code> $\in$ {mod, mod-severe, severe, very severe}	—	3.6% / 4.1%
4	Aortic regurgitation	moderate-or-greater	<code>aortic_regurg</code> $\geq$ moderate (mild-mod = negative)	—	1.4% / 1.2%
5	Mitral regurgitation	moderate-or-greater	<code>mitral_regurg</code> $\geq$ moderate (mild-mod = negative)	—	8.7% / 8.5%
6	Tricuspid regurgitation	moderate-or-greater	<code>tricuspid_regurg</code> $\geq$ moderate (incl. torrential)	—	10.8% / 10.7%
7	Pulmonic regurgitation	moderate-or-greater	<code>pulm_regurg</code> $\in$ {mod/severe, severe, significant}	—	1.7% / 0.8%
8	RV systolic dysfunction	categorical moderate/severe	categorical rv_function moderate/severe (incl. "depressed")	+ TAPSE < 1.7 cm	6.1% / 12.2%
9	Pericardial effusion	moderate or large	moderate / moderate-large / large (small & below = negative)	tamponade descriptors	0.6% / 2.0%
10	Elevated PASP	numeric PASP ≥ 45 mmHg	TR gradient + IVC-derived RAP ≥ 45 ²	fixed RAP 3/5/10 mmHg	10.9% / 15.0%
11	Elevated TR Vmax	direct TR Vmax ≥ 3.2 m/s	direct tr_velocity ≥ 3.2 m/s	derived gradient <code>tr_mmhg</code> ≥ 40.96	12.8% / 10.2%
12	Any SHD (composite)	any of 1–11	GREATEST(1–11); model output index 11	—	47.3% / 52.2%

¹ MIMIC-IV-ECHO provides septal and inferolateral LV wall thickness but **no distinct numeric posterior-wall/LVPW field**; inferolateral is anatomically adjacent but not identical to the posterior wall. ² MIMIC-IV-ECHO exposes **no single numeric PASP field**; RAP estimated from the structured IVC field per ASE categories (0→5→3, 5→10→8, 10→15→13, >15/dilated→20 mmHg); missing/non-visualized IVC → RAP = 3 mmHg.

Inclusion (EchoNext): echo must report LVEF AND ≥ 1 valve finding. **Valve severity logic:** moderate / mod-severe / severe = positive; trace / mild / physiologic = negative; mild-moderate (1–2+) = negative; unquantified "present / can't qualify" = negative. **Exclusions:** age < 18 ; missing age/sex; ventricular-paced; poor-quality/lead-reversal/artifact; all ECG measurements missing; **echocardiograms with repaired or replaced (prosthetic) heart valves** — matching EchoNext/Nature, which excluded these (identified via valve-replacement-structure and leaflet fields).